



STATISTICS SHORT NOTES

For Data Analytics + Data Science + AI/ML

1. Statistics kya hai?

Statistics = Data ko **collect, summarize, analyze aur interpret** karke useful conclusions nikalna.

Data → Statistics → Insight → Decision

2. Types of Data

Numerical

Numbers ke form mein data.

- **Discrete:** countable → number of customers
- **Continuous:** measurable → height, weight, temperature

Categorical

Categories/labels.

- Gender
 - City
 - Product type
-

3. Population vs Sample

Population: Complete group.

Sample: Population ka selected part.

Example:

India ke saare customers = Population

10,000 selected customers = Sample

Data Science mein usually sample se population ke baare mein inference kiya jata hai.

4. Mean, Median & Mode

Mean

Average:

$$\text{Mean} = \Sigma x / n$$

Outliers se strongly affected.

Median

Sorted data ka middle value.

Outliers/skewed data mein useful.

Mode

Most frequently occurring value.

Quick rule:

Normal data → Mean

Outliers/skewed data → Median

Most common category → Mode

5. Range, Variance & Standard Deviation

Range

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

Variance

Mean se observations kitna spread hain, uska squared measure.

Standard Deviation

$$\text{SD} = \sqrt{\text{Variance}}$$

SD jitna high → data utna spread out.

SD jitna low → values mean ke closer.

Practical use: Data variability samajhna, outliers identify karna, distributions compare karna.

6. Percentile & Quartiles

Percentile

Batata hai ki ek value data mein kis position par hai.

Example:

90th percentile salary = ₹80,000

Approximately 90% observations ₹80,000 ya usse kam hain.

Quartiles

Q1 = 25th percentile

Q2 = 50th percentile = Median

Q3 = 75th percentile

7. IQR

Interquartile Range:

$IQR = Q3 - Q1$

Outlier detection:

$Lower = Q1 - 1.5 \times IQR$

$Upper = Q3 + 1.5 \times IQR$

Outside values = **potential outliers**

⚠ Outlier ko automatically delete nahi karna chahiye.

8. Outliers

Outlier = unusual/extreme observation.

Example:

20, 21, 22, 23, 500

500 likely outlier hai.

Outlier handle karne se pehle check:

- Data-entry error?
- Genuine observation?
- Business mein important?
- Model ko affect kar raha hai?

Possible solutions:

- Remove
- Cap

- Transform
 - Robust methods
 - Keep
-

9. Distribution

Distribution batata hai ki data kis pattern mein spread hai.

Common distributions:

- Normal
 - Bernoulli
 - Binomial
 - Poisson
 - Uniform
-

10. Normal Distribution

Bell-shaped distribution.

Normally:

Mean $\approx$ Median $\approx$ Mode

68–95–99.7 Rule

Approximately:

68% $\rightarrow$ Mean $\pm$ 1 SD

95% $\rightarrow$ Mean $\pm$ 2 SD

99.7% $\rightarrow$ Mean $\pm$ 3 SD

Normal distribution statistics aur ML mein bahut important hai.

11. Skewness

Skewness batata hai distribution symmetric hai ya kisi side tilted hai.

Positive / Right Skew

Long tail right side.

Usually:

Mean $>$ Median

Examples:

- Income
- House prices
- Transaction values

Negative / Left Skew

Long tail left side.

Usually:

$$\text{Mean} < \text{Median}$$

12. Z-Score

Z-score batata hai ki value mean se kitne standard deviations door hai.

$$Z = (X - \text{Mean}) / \text{SD}$$

Example:

Mean = 50

SD = 10

X = 70

$$Z = 2$$

Meaning:

70 mean se 2 SD above hai.

Uses:

- Outlier detection
 - Standardization
 - Comparing values
-

13. Probability

Probability = Event hone ki likelihood.

$$0 \leq P(A) \leq 1$$

0 = Impossible

1 = Certain

Basic:

$$P(A) = \text{Favorable outcomes} / \text{Total outcomes}$$

14. Conditional Probability

B hone ke baad A ki probability:

$$P(A|B) = P(A \cap B) / P(B)$$

Used in:

- Classification
 - Fraud detection
 - Recommendation
 - Bayesian methods
-

15. Bayes' Theorem

$$P(A|B) = P(B|A) \times P(A) / P(B)$$

Simple meaning:

New evidence milne ke baad probability update karna.

Used in:

- Naive Bayes
 - Spam detection
 - Medical prediction
 - Risk analysis
-

16. Random Variable

Random experiment ke outcome ko numerical value dena.

Discrete

Countable:

$$\text{Number of customers} = 50$$

Continuous

Measured:

$$\text{Weight} = 65.5 \text{ kg}$$

17. Expected Value

Long-run average expected outcome.

$$E(X) = \sum xP(x)$$

Example:

₹100 milne ki probability = 0.5

₹0 milne ki probability = 0.5

$$E(X) = 100 \times 0.5 + 0 \times 0.5$$

$$= ₹50$$

Useful in:

- Risk
 - Finance
 - Decision making
-

18. Bernoulli & Binomial

Bernoulli

Only two outcomes:

$$\text{Success} = 1$$

$$\text{Failure} = 0$$

Example:

Customer buys / doesn't buy.

Binomial

Multiple Bernoulli trials.

Example:

100 customers mein kitne purchase karenge?

19. Poisson Distribution

Fixed time/space interval mein events ki count.

Examples:

- Calls per hour
 - Website errors per minute
 - Customers arriving per hour
-

20. Sampling

Population se sample lena.

Common methods:

- **Simple Random:** Everyone has equal chance
- **Stratified:** Groups bana kar sample
- **Systematic:** Every kth observation
- **Cluster:** Groups/clusters select karna

⚠ Sample representative hona important hai.

21. Sampling Bias

Agar sample population ko properly represent nahi karta → Sampling Bias.

Examples:

- Selection bias
- Non-response bias
- Survivorship bias

Bad sample → Bad conclusion.

22. Central Limit Theorem (CLT)

Important concept:

Large enough random samples ke sample means ki distribution approximately normal ho jaati hai, even if original population normal na ho.

CLT important hai because it supports:

- Confidence intervals
 - Hypothesis testing
 - Statistical inference
-

23. Standard Error

Standard Error sample statistic ki variability batata hai.

For sample mean:

$$SE = SD / \sqrt{n}$$

Sample size increase:

n ↑ → SE ↓

Meaning:

Larger sample → generally more precise estimate.

24. Confidence Interval

Population parameter ka plausible range estimate.

General formula:

Estimate $\pm$ Critical Value $\times$ SE

Example:

Mean salary = ₹50,000

95% CI:

₹48,000 – ₹52,000

Correct interpretation:

Repeated sampling method ke under, 95% confidence intervals mein approximately 95% true population parameter ko contain karenge.

25. Hypothesis Testing

Statistical claim test karne ke liye.

Null Hypothesis — H_0

Usually:

No difference / No effect

Alternative Hypothesis — H_1

Difference / Effect exists

Example:

H_0 : Old website = New website

H_1 : Old website $\neq$ New website

26. p-value

p-value:

Agar H_0 true ho, toh observed result ya usse more extreme result milne ki probability.

Common significance level:

$$\alpha = 0.05$$

Decision:

$$p < 0.05 \rightarrow \text{Reject } H_0$$

$$p \geq 0.05 \rightarrow \text{Fail to reject } H_0$$

! p-value H_0 true hone ki probability nahi hai.

27. Type I & Type II Error

Type I Error

True H_0 ko reject karna.

False Positive

Type II Error

False H_0 ko reject na karna.

False Negative

Memory:

Type I $\rightarrow$ False Alarm

Type II $\rightarrow$ Missed Detection

28. Statistical Power

Power = True effect hone par us effect ko detect karne ki probability.

$$\text{Power} = 1 - \beta$$

Power increase karne mein help karte hain:

- Larger sample
 - Larger effect
 - Lower variability
-

29. Statistical Tests

t-test

Means compare karne ke liye.

- One-sample t-test
- Independent two-sample t-test
- Paired t-test

Chi-Square Test

Categorical variables ke association ke liye.

Example:

Gender ↔ Product Purchase

ANOVA

3 ya more groups ke means compare karne ke liye.

Example:

Delhi vs Mumbai vs Bangalore sales

30. Covariance

Do variables kis direction mein move karte hain.

Positive → Together increase

Negative → Opposite direction

Problem:

Covariance scale-dependent hai.

Isliye correlation easier to interpret hota hai.

31. Correlation

Variables ke beech linear relationship ki strength aur direction.

$-1 \leq r \leq +1$

+1 → Perfect positive

0 → No linear relationship

-1 → Perfect negative

Pearson

Linear relationship ke liye.

Spearman

Rank/monotonic relationship ke liye.

⚠️ Correlation ≠ Causation

Example:

Ice cream sales ↑ aur drowning cases ↑ ho sakte hain, but ice cream drowning cause nahi karta.

32. Regression

Variables ke relationship ko model karna aur prediction karna.

Simple Linear Regression:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

Where:

y = Target

x = Feature

β_0 = Intercept

β_1 = Coefficient

ε = Error

Example:

Experience → Salary

33. R^2

R-squared batata hai model target ki observed variation ka kitna portion explain karta hai.

Range:

$$0 \rightarrow 1$$

Example:

$$R^2 = 0.80$$

→ Model approximately 80% observed target variation explain karta hai, model definition ke according.

⚠️ High R^2 automatically good model nahi hota.

34. Residual

$$\text{Residual} = \text{Actual} - \text{Predicted}$$

Example:

Actual = 100

Predicted = 90

$$\text{Residual} = 10$$

Residual analysis se pata chal sakta hai:

- Model fit kaisa hai
 - Outliers
 - Non-linearity
 - Unequal variance
-

35. Multicollinearity

Jab independent features ek-dusre se highly correlated hote hain.

Example:

Age ↔ Years of Experience

Problems:

- Regression coefficients unstable
- Interpretation difficult

Common measure:

VIF — Variance Inflation Factor

36. Bias vs Variance

High Bias

Model too simple.

→ **Underfitting**

High Variance

Model training data ko over-learn karta hai.

→ **Overfitting**

Goal:

37. Overfitting & Underfitting

Overfitting

Training performance → High

Test performance → Poor

Solutions:

- More data
- Regularization
- Simpler model
- Cross-validation
- Feature selection

Underfitting

Training → Poor

Test → Poor

Solutions:

- Better features
- More suitable model
- More complexity
- Less excessive regularization

38. Standardization

Feature ko approximately:

Mean = 0

SD = 1

mein convert karta hai.

Formula:

$$z = (x - \mu) / \sigma$$

Useful for:

- Logistic Regression
- KNN

- SVM
 - Neural Networks
 - PCA
-

39. Normalization

Min-Max scaling:

$$x' = (x - \min) / (\max - \min)$$

Usually values:

$$0 \rightarrow 1$$

⚠ Standardization aur normalization same nahi hain.

40. EDA — Exploratory Data Analysis

Data analysis ka first major step.

Basic workflow:

Understand Data

↓

Missing Values

↓

Duplicates

↓

Data Types

↓

Descriptive Statistics

↓

Distribution

↓

Outliers

↓

Correlation

↓

Visualization



Business Insights

Useful Python:

```
df.info()
df.describe()
df.isnull().sum()
df.corr(numeric_only=True)
```

41. A/B Testing

Do versions compare karna.

Example:

A = Old Website

B = New Website

Check:

Kya B ki conversion genuinely better hai ya difference random variation ki wajah se hai?

Basic process:

Define Metric → H_0/H_1 → Experiment → Statistical Test → p-value/CI
→ Decision

42. Effect Size

p-value sirf statistical evidence batata hai.

Effect size batata hai:

Difference kitna bada hai?

Important because:

Statistically significant ≠ Practically important

Example:

10.00% → 10.01%

Statistically significant ho sakta hai, but business impact tiny ho sakta hai.

43. Missing Data

Common approaches:

- Remove rows
- Remove columns
- Mean imputation
- Median imputation
- Mode imputation
- Advanced/model-based imputation

⚠ Blindly missing values ko mean se replace mat karo.

44. Data Leakage

ML mein data leakage:

Training ke time aisi information use karna jo real prediction time par available nahi hogi.

Result:

Training performance → Unrealistically high

Real-world performance → Poor

Data leakage ML projects ka major mistake hai.

45. Cross-Validation

Model ko multiple train/validation splits par evaluate karna.

Common:

K-Fold Cross Validation

Example:

K = 5

Data ko 5 parts mein divide karke model ko different folds par validate karte hain.

Benefits:

- Better performance estimate
 - Limited data ka better use
 - Model comparison
-

46. Classification Metrics

Accuracy

$\text{Correct Predictions} / \text{Total Predictions}$

Precision

$\text{TP} / (\text{TP} + \text{FP})$

Question:

Predicted positive mein se kitne actually positive?

Recall

$\text{TP} / (\text{TP} + \text{FN})$

Question:

Actual positive mein se kitne detect hue?

F1 Score

$2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$

Precision + Recall ka balance.

47. Confusion Matrix

Four terms:

TP = True Positive

TN = True Negative

FP = False Positive

FN = False Negative

Memory:

True/False = Prediction correct/wrong

Positive/Negative = Predicted class

48. Precision vs Recall

Precision important

Jab false positives costly hain.

Example:

Recall important

Jab false negatives costly hain.

Example:

Fraud detection / Disease screening

49. ROC-AUC

ROC curve:

True Positive Rate vs False Positive Rate

AUC:

0.5 → Random-like

1.0 → Perfect

Higher AUC generally better.

⚠ Imbalanced data mein Precision-Recall curve bhi important hai.



MASTER FORMULA CHEAT SHEET

Central Tendency

$$\text{Mean} = \Sigma x/n$$

Range

$$\text{Range} = \text{Max} - \text{Min}$$

IQR

$$\text{IQR} = Q3 - Q1$$

Z-score

$$Z = (X - \mu)/\sigma$$

Conditional Probability

$$P(A|B) = P(A \cap B)/P(B)$$

Bayes

$$P(A|B) = P(B|A)P(A)/P(B)$$

Expected Value

$$E(X) = \sum xP(x)$$

Standard Error

$$SE = SD/\sqrt{n}$$

Confidence Interval

$$\text{Estimate} \pm \text{Critical Value} \times SE$$

Regression

$$y = \beta_0 + \beta_1 x + \varepsilon$$

Precision

$$TP / (TP + FP)$$

Recall

$$TP / (TP + FN)$$

F1

$$2PR / (P + R)$$

WHAT TO MASTER FIRST

MUST KNOW

1. Mean
2. Median
3. Standard Deviation
4. Variance
5. Percentile
6. IQR
7. Outliers
8. Normal Distribution
9. Z-score
10. Probability
11. Conditional Probability
12. Bayes
13. Sampling
14. CLT
15. Standard Error

- 16. Confidence Interval
 - 17. Hypothesis Testing
 - 18. p-value
 - 19. t-test
 - 20. Chi-square
 - 21. ANOVA
 - 22. Correlation
 - 23. Regression
 - 24. R^2
 - 25. Multicollinearity
 - 26. Bias-Variance
 - 27. Standardization
 - 28. EDA
 - 29. Cross-validation
 - 30. Classification Metrics
-

ONE-LINE REVISION

Mean/Median → Center

SD/Variance → Spread

Percentile/IQR → Position & spread

Outlier → Unusual value

Distribution → Data ka pattern

Z-score → Mean se kitni SD door

Probability → Chance

Bayes → Evidence se probability update

CLT → Sample means approximately normal

SE → Estimate ki uncertainty

CI → Plausible parameter range

p-value → H_0 ke against evidence

t-test → Means compare

Chi-square → Categorical association

ANOVA → 3+ means compare

Correlation → Linear relationship

Regression → Relationship + prediction

R² → Explained variation

Multicollinearity → Features highly related

Bias → Underfitting

Variance → Overfitting

Scaling → Features ko comparable scale par lana

EDA → Data ko understand karna

Cross-validation → Model reliability check

Precision → Predicted positives mein kitne correct

Recall → Actual positives mein kitne found